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# Expectation-Maximization (EM) Algorithm
# Using Gaussian Mixture Model

import numpy as np
import matplotlib.pyplot as plt
from sklearn.mixture import GaussianMixture

# Create sample data
X = np.array([
    [1, 2], [1.5, 1.8], [2, 2.2], [2.5, 2],
    [8, 8], [8.5, 8.2], [9, 8.5], [9.5, 9]
])

# Create Gaussian Mixture Model
model = GaussianMixture(
    n_components=2,
    random_state=42
)

# Apply EM algorithm
model.fit(X)

# Predict cluster labels
labels = model.predict(X)

# Print results
print("Expectation-Maximization Algorithm")
print("-----------------------------------")

print("\nCluster Labels:")
print(labels)
```

```python
print("\nMeans of Gaussian Components:")
print(model.means_)

print("\nCovariance Matrices:")
print(model.covariances_)

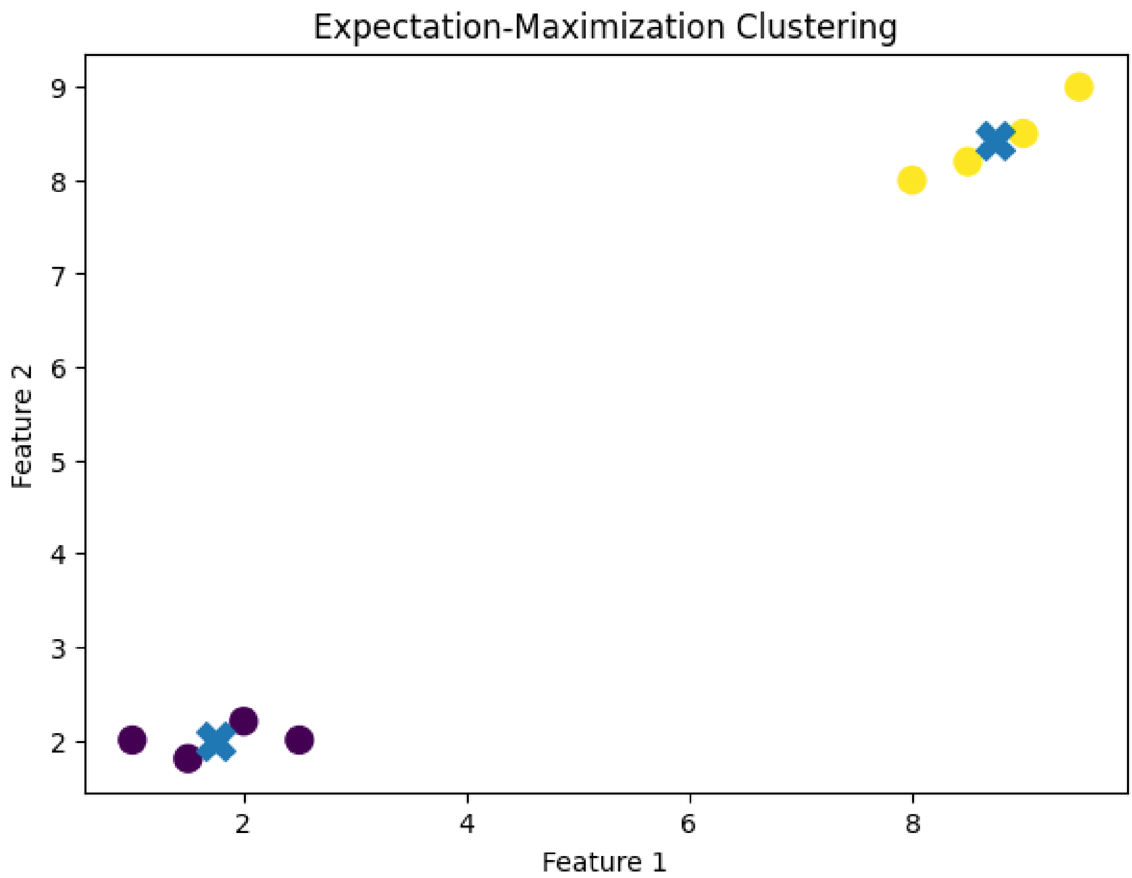
print("\nWeights of Components:")
print(model.weights_)

# Plot the clusters
plt.figure(figsize=(7, 5))

plt.scatter(
    X[:, 0],
    X[:, 1],
    c=labels,
    s=100
)

plt.scatter(
    model.means_[:, 0],
    model.means_[:, 1],
    marker='X',
    s=200
)

plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("Expectation-Maximization Clustering")
plt.show()
```

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Expectation-Maximization Algorithm
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Cluster Labels:
[0 0 0 0 1 1 1 1]

Means of Gaussian Components:
[[1.75  2.   ]
 [8.75  8.425]]

Covariance Matrices:
[[[0.312501 0.025   ]
  [0.025    0.020001]]

 [[0.312501 0.20625 ]
  [0.20625  0.141876]]]

Weights of Components:
[0.5 0.5]
```